PRACTICAL-7

Aim
To implement a simple probabilistic classifier for diagnosis using the Naive Bayes algorithm and perform probabilistic inference using Python.

Program:

```
# Practical 7: Simple Probabilistic Classifier for Diagnosis

from sklearn.naive_bayes import BernoulliNB

# Training dataset
# [fever, cough, fatigue]
# 1 = Yes, 0 = No

X = [
    [1, 1, 1],
    [1, 1, 0],
    [1, 0, 1],
    [0, 1, 0],
    [0, 1, 1],
    [0, 0, 0]
]

# Diagnosis
y = [
    'Flu',
    'Flu',
    'Flu',
    'Cold',
    'Cold',
    'Cold'
]

# Create and train Naive Bayes model
model = BernoulliNB()
model.fit(X, y)
```

```
# Predict diagnosis for a new patient
patient = [[1, 1, 1]]

prediction = model.predict(patient)

print("Predicted diagnosis:", prediction[0])

# Get probability of each diagnosis
probabilities = model.predict_proba(patient)

for diagnosis, probability in zip(model.classes_, probabilities[0]):
    print(diagnosis, "=", round(probability, 2))

# Test another patient
patient2 = [[0, 1, 0]]

prediction2 = model.predict(patient2)

print("Predicted diagnosis for patient 2:", prediction2[0])
```

output:

```
Predicted diagnosis: Flu
Cold = 0.14
Flu = 0.86
Predicted diagnosis for patient 2: Cold
```